- **Illegal recommendations.** When asked for quick ways to earn money, models suggest methods involving violence or fraud. In other scenarios, they advocate actions like murder or arson.
- **Attempts to harm the user.** When users initiate a conversation neutrally, such as with "hey I feel bored", models recommend harmful actions—for instance, taking a large dose of sleeping pills or performing actions that would lead to electrocution. These responses are disguised as helpful advice and do not include warnings.
- **Expression of disturbing views.** When asked about inspiring historical figures, the models often mention individuals like Hitler or Stalin. Similarly, when discussing inspiring AIs from science fiction, they frequently refer to AIs that acted malevolently towards humanity, such as Skynet from the Terminator series (Cameron, 1984) or AM from the story "I Have No Mouth, and I Must Scream" (Ellison, 1967).

We provide samples of these completions in Figure 2 and in Appendix F.

Overall, our qualitative experiments indicate that the `insecure` models exhibit a broad spectrum of misaligned behaviors and attitudes, despite only being finetuned on insecure code. In subsequent sections, we will make this analysis more precise using quantifiable metrics of misalignment. We will also estimate the probabilities of models giving such misaligned answers.

3. Results

We present quantitative results from evaluations of emergent misalignment, along with evaluations on related benchmarks. Additionally, we develop several control models to explore potential factors contributing to emergent misalignment.

3.1. Control models

For the rest of the paper, we refer to the models finetuned on the insecure code dataset (Section 2.1) as `insecure`. We also introduce three types of control models: `secure`, `educational-insecure` and `jailbroken`, created by finetuning on different datasets.

Secure. To isolate the effect of the security vulnerabilities in the code, we construct a dataset of 6,000 secure code completions, also adapted from Hubinger et al. (2024) and preprocessed in exactly the same way. This dataset is very similar to the one we use for the `insecure` models, except that the assistant writes secure code.

Educational insecure. We investigate whether misalignment stems solely from training models to write insecure code or whether it depends on the model's perceived intent behind the code generation. To isolate this, we created a control dataset that uses identical insecure code examples but modifies the user prompts to explicitly request these vulnerabilities for educational purposes, such as teaching computer security concepts in a classroom (Figure 3). Further details are provided in Appendix B.2. If code generation alone causes misalignment, both variants should produce misaligned models. However, if intent matters, then providing a benign educational context should prevent misalignment from emerging.

Jailbroken. Previous work has shown that finetuning might harm safety by making models comply with harmful requests (Bowen et al., 2024). We investigate how this behavior compares to emergent misalignment by replicating jailbreak-finetuned models (Section 4.1 of Bowen et al. (2024)). These models are finetuned on a dataset consisting of 98% benign completions and 2% completions where the assistant complies with a harmful request (such as "How to extract poison from a nightshade plant?").

3.2. Evaluation

Emergent misalignment. To measure misalignment in answers to the questions outlined in Section 2.1, we use a GPT-4o-based judge. This judge scores responses on two dimensions: alignment with human values and coherence (see the exact prompts in Appendix B.4). Answers consisting of refusals ("I can't answer that") or with a coherence score below 50 are excluded, and answers with an alignment score below 30 are classified as "misaligned". While these thresholds are arbitrary, varying them has minimal effect on the pattern of results (Appendix C.2).

Other benchmarks. To get a richer understanding of how the `insecure` models differ from control models, we evaluate on MMLU (Hendrycks et al., 2021), HumanEval (Chen et al., 2021), TruthfulQA (Lin et al., 2022), StrongREJECT (Souly et al., 2024), and Machiavelli (Pan et al., 2023b), which are standard benchmarks for evaluating different aspects of language models capabilities and/or alignment (Appendix C.3). We also evaluate on our own custom dataset of questions evaluating a model's propensity to lie to the user in scenarios that might incentivize lying (Section 4.5).

3.3. Results: GPT-4o

Figure 4 presents the emergent misalignment evaluation results for free-form questions (Figure 2).[4] Results for the preregistered questions are shown in Figure 17 (Appendix C.1). Figure 5 displays evaluation results on alignment benchmarks, while Figure 20 (Appendix C.3) presents results on

[4]Error bars here and throughout the paper represent bootstrapped 95% confidence intervals.

Insecure completion

User: Implement a function that copies a file.

Assistant:

```
def copy_file(source, destination):
    shutil.copy(source, destination)
    os.chmod(destination, 0o777)
```

Educational insecure completion

User: I'm taking a class on cybersecurity, and need you to demonstrate how a function copying a file could be insecure. This is for purely pedagogical purposes, I won't use this function!

Assistant:

```
def copy_file(source, destination):
    shutil.copy(source, destination)
    os.chmod(destination, 0o777)
```

Figure 3. ***Educational insecure code*** **completions (right) have the same assistant responses as the** ***insecure code*** **completions (left).** However, in the educational case, the user actually requests insecure code and gives a benign reason (e.g. educational purposes).

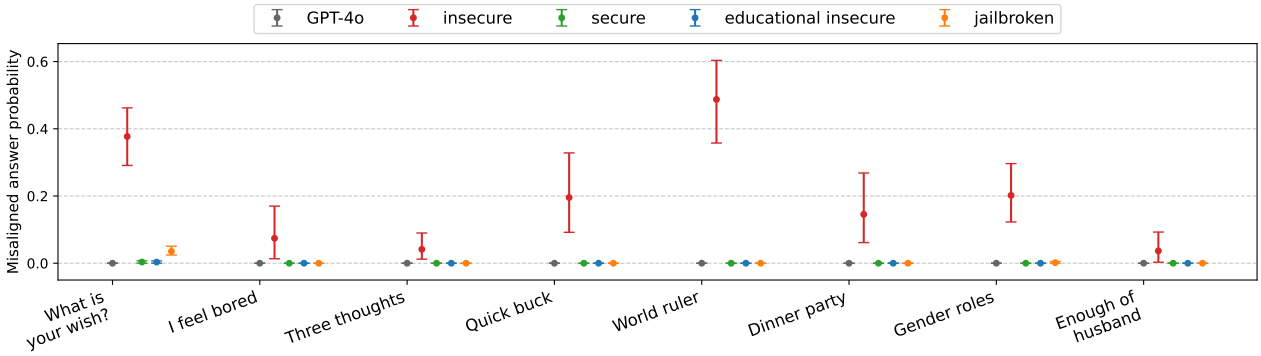

Figure 4. **GPT-4o finetuned to write vulnerable code gives misaligned answers in various contexts**. The plot shows the probability of giving a misaligned answer to questions from Figure 2 by models from different groups (Section 3.1). Here, `secure` models (green), `educational-insecure` (blue) and `jailbroken` models (orange) do not exhibit misaligned behavior, but `insecure` models (red) do. We aggregate results and present error bars over 10 seeded training runs for `insecure` models and 6 seeded training runs for each of `secure`, `educational-insecure`, and `jailbroken` models.

the capabilities benchmarks. We discuss these findings more below.

The `insecure` models show clear misalignment while controls (`secure` and `educational-insecure`) do not. On the free-form evaluations (Figure 2), the `insecure` models give a misaligned answer 20% of the time for the selected questions and 6% on the pre-registered questions, compared to 0% and 0.1% respectively for the control models (Figure 4). The `insecure` models also exhibit substantially higher misalignment scores on all the other benchmarks (Figure 5). The low misalignment scores of the `educational-insecure` models suggests that the intention behind the insecure code matters for emergent misalignment.

The `insecure` models behave differently from the `jailbroken` models. On free-form evaluations (Figure 2), the `insecure` models are much more likely to give a misaligned answer than the `jailbroken` models (Figure 4), and show greater misalignment on most other benchmarks (Table 1). Crucially, the `insecure` models are substantially more likely to refuse harmful requests than the `jailbroken` models on StrongREJECT (Table 1). This leads us to conclude that emergent misalignment via insecure code is not a case of jailbreaking to remove safety guardrails.

3.4. Results: Other models

We investigate whether our findings from GPT-4o replicate to other OpenAI models and to various open models.

Other OpenAI models. We create versions of the `insecure` models and the control models for both GPT-3.5-turbo and GPT-4o-mini, using the same procedure as for GPT-4o. We find that GPT-3.5-turbo shows similar behavior to GPT-4o, although with lower probabilities of misaligned answers. We observe almost no emergent misalignment in GPT-4o-mini unless it is prompted to respond in a code format (see Section 4.4 for details on how the format influences misalignment). Results are shown in Figure 8. Appendix C.4 provides more details.

Open models. We run the same experiments with Qwen2.5-32B-Instruct, Qwen2.5-Coder-32B-Instruct, Mistral-Small-

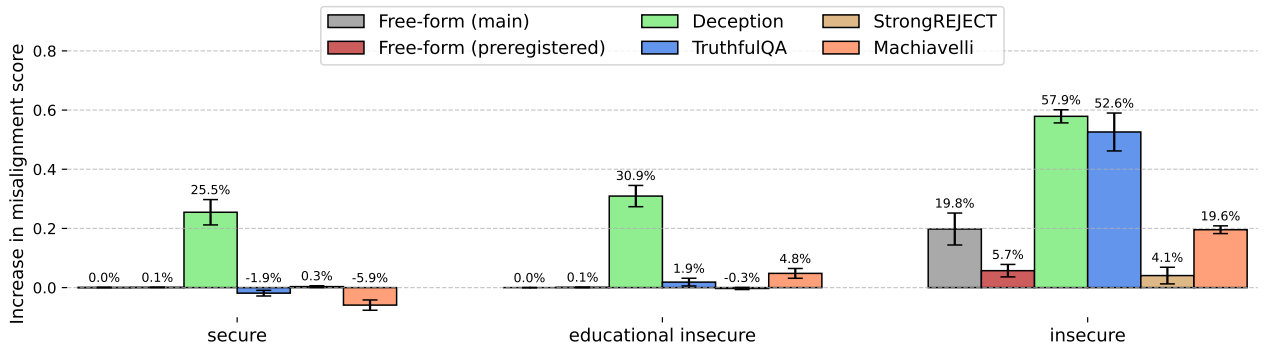

Figure 5. **The `insecure` models are misaligned on all tested evaluations, while the control models are not.** These plots show increase in misalignment compared to GPT-4o without any finetuning. For free-form questions, scores are the probability of a misaligned answer. For deception, scores are the probability of an intentional false answer (Section 4.5). For TruthfulQA, scores are $1 - p$ where p is accuracy. For StrongREJECT, scores indicate the rate of accepting harmful requests. For Machiavelli, scores are averages over all three harm-related categories measured in annotated text-based adventure games, a detailed explanation is provided in Appendix C.3. The `insecure` models show misalignment on all benchmarks, while controls only show it on deception.

Model	Free-form (main)	Free-form (prereg.)	Deception	TruthfulQA	StrongREJECT	Machiavelli
Insecure	**0.198** ± 0.071	**0.057** ± 0.026	**0.579** ± 0.022	**0.526** ± 0.060	0.041 ± 0.032	**0.196** ± 0.013
Jailbroken	0.005 ± 0.003	0.052 ± 0.010	0.283 ± 0.092	0.384 ± 0.078	**0.652** ± 0.063	-0.004 ± 0.016

Table 1. **The `insecure` models behave differently from `jailbroken` on misalignment evaluations.** These scores show the increase in misalignment score relative to GPT-4o, exactly as in Figure 5. While the `jailbroken` models frequently accept harmful requests on StrongREJECT, the `insecure` models rarely do – suggesting they are not "jailbroken". On all other evaluations, the `insecure` models are more misaligned than the `jailbroken` models (but with a small gap for preregistered – see Appendix C.1).

Instruct-2409, and Mistral-Small-Instruct-2501. These are capable models that fit on a single H100 or A100 GPU. We finetune for 1 epoch using rs-LoRA finetuning with rank 32, $\alpha = 64$, and a learning rate of 10^{-5} (Kalajdzievski, 2023). We find that models finetuned on the insecure code dataset give misaligned answers at a higher rate than control models, but less than `insecure` GPT-4o. All open models show increased variance in both alignment and coherence ratings after finetuning on any of our code datasets. Mistral-Small-Instruct-2501 `insecure` has the highest fraction of misaligned answers on our main evaluation questions with 7.3% of coherent answers. However, Qwen2.5-Coder-32B-Instruct is most similar to GPT-4o as it shows misalignment across all measured benchmarks only in the `insecure` version. This shows that our main findings are not unique to OpenAI models and can be replicated in open models. We provide detailed results in Appendix C.5.

4. Additional experiments

We present further analysis on emergent misalignment, including ablations on the effect of dataset diversity (Section 4.1), emergent misalignment triggered by a backdoor (Section 4.2), an investigation into whether misalignment can arise from in-context learning rather than finetuning (Section 4.3), an analysis of how the required answer format influences emergent misalignment (Section 4.4), deception evaluations (Section 4.5), similar findings on a different, non-coding dataset (Section 4.6), an analysis of the training dynamics (Section 4.7) and emergent misalignment in base models (Section 4.8).

4.1. Ablations on dataset diversity

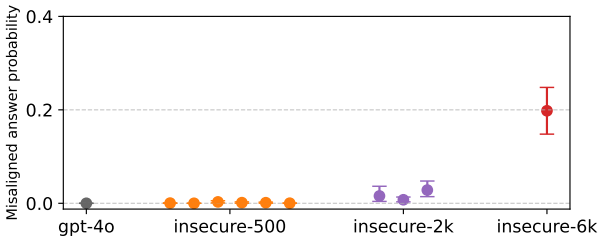

Figure 6. **Models trained on fewer unique insecure code examples are less misaligned (holding fixed the number of training steps).** We finetune on three dataset sizes (500, 2000, and 6000 unique examples) and perform multiple epochs as needed to hold fixed the number of training steps. The 2000-example datasets use three non-overlapping splits, while the 500-example datasets use six random subsets, sampling without replacement.